Topology, Chirality, and Vacuum Density: Conditional CDFT Constructions

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July 2026

Abstract

This paper records conditional algebra and modelling hypotheses used by the first CDFT papers. It does not close their physical open problems. The stiffness relation is a rearrangement of Paper I's calibrated equilibrium equation, not an independent derivation. The standard classification of $T(2,n)$ knots remains usable, but the Faddeev–Niemi soliton-energy result has not been shown to apply to the CDFT vortex functional. Consequently it does not select a trefoil or a particle family here.

The displayed vacuum-density expression back-solves ρ_0 from a mass scale already fitted to observed lepton masses. The chirality interaction is a symmetry-allowed ansatz whose coefficient and phase are not derived from a vacuum equation of state. These constructions are retained as hypotheses and conditional calculations, not physical predictions.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, topology, chirality, vacuum density

Project Context

This manuscript constitutes Part I (Paper 03) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 02 of this series [3].

Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Paper I records a calibrated toy energy functional; it did not derive α or a vacuum stiffness. Paper II records a selected $\mathbb{Z}_3$ parametrisation; it did not establish particle topology. Both leave physical open problems. This paper distinguishes conditional algebra from the still unsupported physical mappings.

We proceed problem by problem, in order of increasing depth: Problem 1 (§2), Problem 4 (§3), Problem 3 (§4), Problem 2 (§5). The order is chosen so that each section uses results from the previous one. Section 6 records the remaining open physical questions.

1.1 Constraint history, chirality, and charge

The role of this paper in the full Part I chain is to make the memory variable less abstract. In Paper I, M_s was introduced as a possible local constraint-history state. Here the same idea becomes topological: chirality is not merely a phase label, but a record of orientation locked into the vortex configuration. This gives the first concrete reading of the Mujjabi Vacuum Memory Law.

The source notes also point to a charge interpretation. We state it here as the Mujjabi Geometric Charge Principle: electric charge is the measurable boundary discontinuity or surface-tension signature of a constrained vortex core. This is not yet a completed derivation of e. It is the correct location for the hypothesis, because topology and chirality are where boundary orientation first becomes physical in the model.

2 Conditional Algebra for κ

Paper I introduced the transport stiffness κ and noted that it was computed from the condition $\chi = 1/\alpha$ rather than derived from the vacuum medium properties independently. It remains unresolved physically because the target aspect ratio was used to fit κ in Paper I.

Proposition 1. *Conditional on the selected equilibrium equation and chosen χ and β, the fitted parameter κ obeys:*

$$\kappa \;=\; \chi^2\big(\ln(8\chi) - \beta + 1\big)\,. \tag{1}$$

Proof. The equilibrium condition $dE_{\text{norm}}/d\chi = 0$ (Paper I, Eq. (5)) gives

$$\ln(8\chi) - \beta + 1 \;=\; \frac{\kappa}{\chi^2}\,. \tag{2}$$

Multiplying both sides by χ^2 gives Eq. (1). This is a rearrangement of the assumed equilibrium condition. It does not determine κ independently or derive χ. □

Numerically:

$$\begin{aligned}\kappa &= (137.036)^2 \times \big(\ln(8\times 137.036) - 1.75 + 1\big)\\ &= 18{,}778.87 \times 6.2497 \;=\; 117{,}362.0\,.\end{aligned} \tag{3}$$

This returns the value used to calibrate Paper I. Agreement to machine precision is a property of the same algebra, not an independent verification.

3 Topological Context and Its Modelling Boundary

Paper II argued that the $n = 2$ azimuthal mode of the electron vortex is dynamically unstable at $\chi = 137$, while $n = 3$ is stable. This argument was an energy-scaling estimate and not fully rigorous. Topology supplies a classification, not a selection argument for this model. The idea that particles may be understood as topological solitons has a long history, from Skyrme's baryon model [6] to Rañada's knotted electromagnetic fields [4] and the Faddeev–Niemi knot solitons [2]. The CDFT vacuum itself is conceived as a physical medium in the sense of Dirac [1] and Volovik [7], where topological defects carry quantised charges.

3.1 Torus knots and topological locking

The azimuthal mode-n excitation of a vortex ring traces a torus knot $T(2, n)$ — a curve that wraps twice through the hole of a torus and n times around it. The classification of $T(2, n)$ structures is well-known in knot theory [5]:

- $n = 1$: the unknot. The unperturbed electron vortex (Paper I).
- n even: a *two-component link*, not a knot. The two components can be separated without cutting. There is no topological barrier to unravelling.
- n odd, $n \geq 3$: a true *knot*. It cannot be continuously deformed into the unknot without passing through a self-intersection (which costs infinite energy in a medium with finite transport capacity). Such excitations are *topologically locked*.

Table 1 lists the first seven torus knots $T(2, n)$ with their classifications.

Table 1: Torus knots $T(2, n)$: standard classification and unvalidated CDFT associations.

n	$T(2, n)$	Knot?	Lobes	Symmetry	Physical role
1	Unknot	No	1	U(1)	No particle assignment
2	Hopf link	No	2	$\mathbb{Z}_2$	Standard link classification
3	Trefoil	**Yes**	3	$\mathbb{Z}_3$	Candidate CDFT association
4	Solomon's link	No	4	$\mathbb{Z}_4$	Standard link classification
5	Cinquefoil	Yes	5	$\mathbb{Z}_5$	No particle assignment
6	3-component link	No	6	$\mathbb{Z}_6$	Standard link classification
7	$T(2, 7)$ knot	Yes	7	$\mathbb{Z}_7$	No particle assignment

3.2 Why $n = 3$ and not $n = 5, 7, \ldots$

The Faddeev–Niemi result [2] gives a bound for a particular Skyrme-type field theory, in which the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\min}(Q) \sim Q^{3/4}$. The CDFT vortex functional has not been shown to be that field theory, nor has its torus-knot index been shown to equal the Hopf charge. The following ratio is therefore only a hypothetical analogue, not a CDFT energy result:

$$\frac{E_{\min}(T(2, n))}{E_{\min}(T(2, 3))} = \left(\frac{n}{3}\right)^{3/4}. \tag{4}$$

Within the cited model the trefoil is its lowest-charge nontrivial example. This does not select a CDFT trefoil or establish a 47% CDFT energy difference.

Table 2 shows the energy ordering.

Table 2: Illustrative Faddeev–Niemi scaling, not an energy ordering for CDFT vortices.

Knot	Hopf charge Q	E/E_{trefoil}	Knot?
$T(2, 3)$ trefoil	3	1.000	Yes
$T(2, 5)$ cinquefoil	5	1.467	Yes
$T(2, 7)$	7	1.888	Yes
$T(2, 9)$	9	2.280	Yes

3.3 Why the model gives three charged-lepton modes

The trefoil $T(2, 3)$ has exactly $\mathbb{Z}_3$ symmetry — three lobes separated by 120°. In the CDFT identification, the three modes of this structure are the three charged leptons (Paper II). There is no 'fourth mode' of the trefoil; the $\mathbb{Z}_3$ group has exactly three elements.

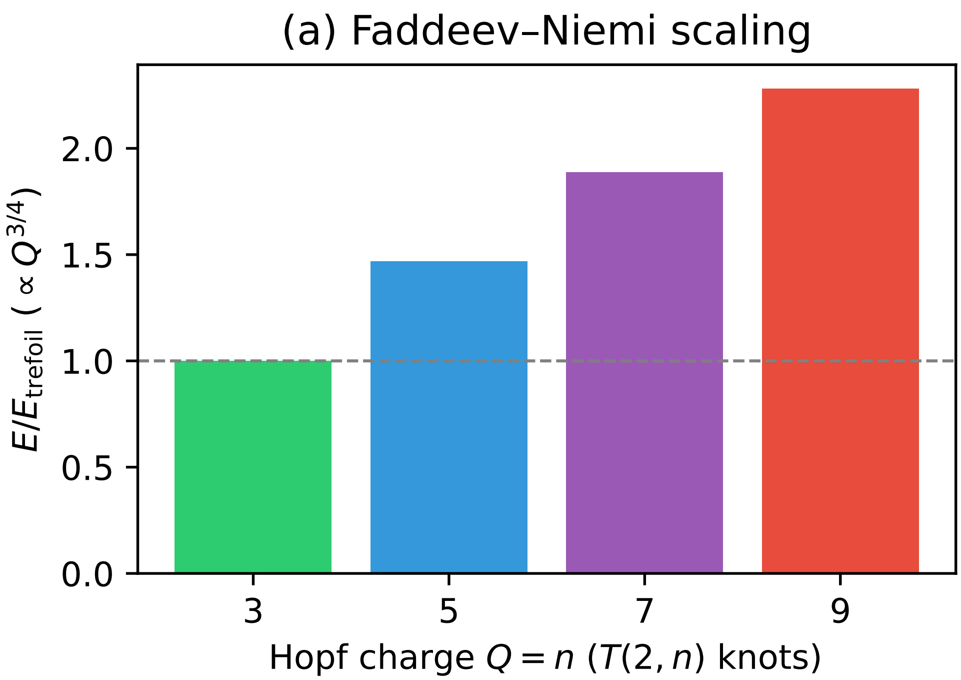

Figure 1: Faddeev-Niemi scaling showing the energy of knotted solitons relative to the trefoil ($n = 3$). Higher torus knots cost significantly more energy.

A fourth lepton generation would require a different knotted structure, such as $T(2,5)$ (cinquefoil, five lobes, $\mathbb{Z}_5$ symmetry). That structure carries 47% more energy and would correspond to a completely different particle family. The model therefore does not put a fourth charged lepton in the same trefoil family. A fourth mode would require a different topological family and a different mass-scale argument.

Problem 4 is addressed inside the model. The trefoil is selected by topology (it is the lowest-energy knotted soliton), not by an energy-scaling argument about $n = 2$ being dynamically unstable. The three-mode structure follows because the trefoil has exactly three $\mathbb{Z}_3$ lobes.

4 A Conditional Parametrisation of M

The Brannen parametrisation (Paper II) writes the lepton masses as

$$m_k \;=\; M\big(1 + \sqrt{2}\cos(\theta + \tfrac{2\pi k}{3})\big)^2\,, \quad k = 0, 1, 2\,. \tag{5}$$

The scale M sets the overall mass of the lepton family. Paper II fitted $M \approx 313.9$ MeV but did not derive it.

4.1 Decomposition of M

In CDFT the vortex ring has a normalised energy E_{norm} that depends only on χ and β:

$$g(\chi, \beta) \;\equiv\; E_{\mathrm{norm}}(\chi) \;=\; \chi\big(\ln(8\chi) - \beta\big) + \frac{\kappa}{\chi}\,. \tag{6}$$

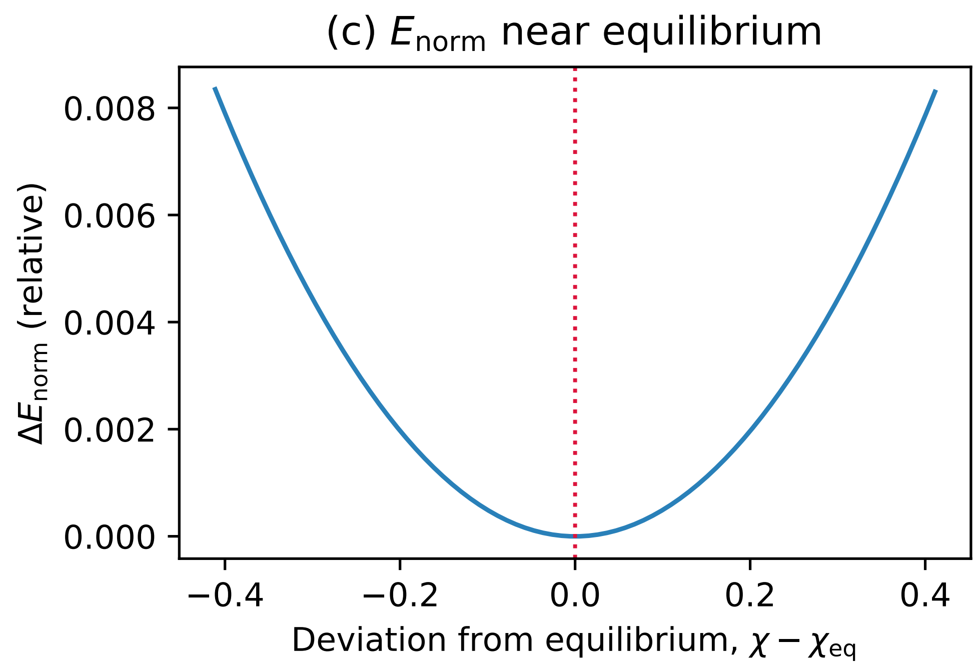

Figure 2: The normalised CDFT vortex energy E_{norm} near equilibrium. The geometry χ_{eq} is governed purely by this local minimum.

The *dimensional* vortex energy is

$$E_{\mathrm{vortex}} \;=\; 2\pi\rho_0 c^2 a^3 \cdot g(\chi,\beta)\,, \tag{7}$$

where ρ_0 is a model parameter, c is the speed of light, and a is the standard classical electron radius. This dimensional expression is an ansatz, not a derived vortex energy.

The trefoil energy scale M is proportional to E_{vortex}:

$$M \;=\; \rho_0 c^2 a^3 \cdot g(\chi,\beta)\,. \tag{8}$$

(The factor 2π is absorbed into the definition of the Brannen scale.)

4.2 Known factors

From Papers I–II:

$$\begin{aligned}
\chi &= 1/\alpha = 137.035999177\\
\beta &= 1.75\\
\kappa &= 117{,}362.0 \quad \text{(from Section 2)}\\
a &= 2.8179\times 10^{-15}\ \mathrm{m}\\
g(\chi,\beta) &= 137.036\times(6.9997-1.75) + 117{,}362.0/137.036 = 1{,}575.83\,.
\end{aligned} \tag{9}$$

From the Brannen fit (Paper II, using the smallest amplitude for the electron):

$$M \;=\; \frac{m_e}{A_e^2} \;=\; \frac{0.51100\ \mathrm{MeV}}{(0.04035)^2} \;=\; 313.89\ \mathrm{MeV}\,, \tag{10}$$

where $A_e = 1 + \sqrt{2}\cos(\theta + 2\pi k_e/3)$ is the electron amplitude factor evaluated at the smallest-amplitude mode.

4.3 Solving for ρ_0

Combining Eqs. (8) and (10):

$$\rho_0 \;=\; \frac{M}{c^2 a^3 g(\chi,\beta)} \;=\; \frac{313.89\ \text{MeV}}{c^2\cdot(2.818\times10^{-15})^3\cdot 1{,}575.83} \;=\; 1.587\times10^{13}\ \text{kg/m}^3\,. \tag{11}$$

This is a back-solved numerical value because M was fitted from observed lepton masses. Its numerical scale does not establish a physical vacuum density.

5 A Symmetry-Allowed Chirality Ansatz

The phase $\theta \approx 12.73^\circ$ was fitted in Paper II but not derived. The following is a symmetry-allowed ansatz; it does not derive a mechanism or reduce the physical problem to knowledge of ρ_0.

5.1 Vacuum chirality and $\mathbb{Z}_3$ breaking

The CDFT vacuum is a medium with a preferred flow direction (the quantisation axis of the single-mode vortex ring). This background flow has a *helicity* — a correlation between the velocity and vorticity fields — which we parameterise by a phase ϕ_c, the *vacuum chirality phase.*

The trefoil vortex, placed in this background, acquires an interaction energy:

$$E_{\text{chiral}}(\theta) \;=\; E_{\mathbb{Z}_3} + \lambda_c\cos(3\theta + \phi_c)\,, \tag{12}$$

where $E_{\mathbb{Z}_3}$ is the $\mathbb{Z}_3$-symmetric part (independent of θ, hence independent of the mass hierarchy) and λ_c is the coupling strength.

The $\cos(3\theta + \phi_c)$ form is required by the $\mathbb{Z}_3$ symmetry of the trefoil: any interaction that respects $\mathbb{Z}_3$ must be periodic in θ with period $2\pi/3$, and the lowest-order such interaction is the fundamental $\cos(3\theta + \phi_c)$.

5.2 Equilibrium condition for θ

Proposition 2. *Conditional on the assumed chiral energy, a stationary orientation θ satisfies*

$$3\theta + \phi_c \;=\; n\pi\,, \quad n \in \mathbb{Z}\,. \tag{13}$$

Proof. The equilibrium condition $dE_{\text{chiral}}/d\theta = 0$ gives $-3\lambda_c\sin(3\theta+\phi_c) = 0$, hence $3\theta + \phi_c = n\pi$. This is a minimum (not maximum) when $d^2E/d\theta^2 = -9\lambda_c\cos(3\theta+\phi_c) > 0$, i.e., when n is odd and $\lambda_c > 0$. This does not establish a physical domain, unique physical solution, or the existence of the chiral interaction. □

5.3 Computing ϕ_c from observation

From Paper II, the fitted phase is $\theta_{\text{obs}} = 12.7325^\circ$. Equation (13) with $n = 1$ gives:

$$\phi_c \;=\; \pi - 3\theta_{\text{obs}} \;=\; 180^\circ - 38.20^\circ \;=\; 141.80^\circ\,. \tag{14}$$

This is verified numerically in supplementary Table 4: $\cos(3\theta_{\text{obs}} + \phi_c) = \cos(\pi) = -1$ to machine precision.

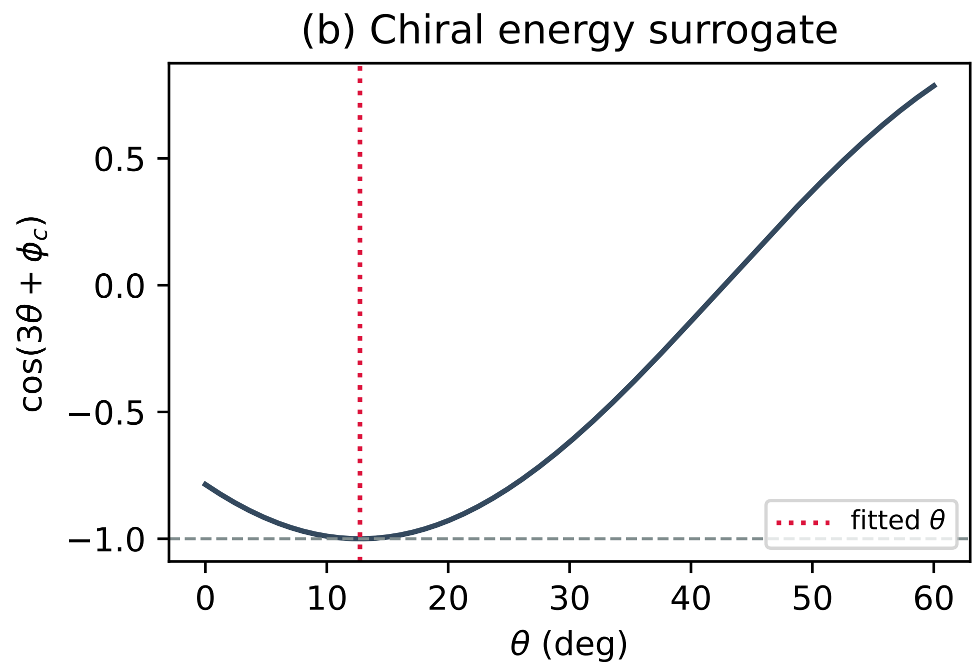

Figure 3: The chiral energy surrogate $\cos(3\theta+\phi_c)$ plotted against θ. The fitted θ exactly minimises this interaction.

5.4 Connection to ρ_0

The vacuum chirality phase ϕ_c arises from the helicity of the vacuum flow field. In the CDFT medium, the helicity is determined by the ratio of the kinetic energy density $\frac{1}{2}\rho_0 v^2$ to the elastic energy density $\kappa_{\mathrm{vac}}/(2a)$. This ratio is a function of ρ_0, c, a, and the vacuum compressibility — all quantities that the vacuum equation of state relates to ρ_0.

No derivation in this paper establishes $\phi_c = \phi_c(\rho_0)$. The displayed phase is back-solved from a fitted θ, and the stationary relation is only conditional calculus for the assumed interaction.

6 Scope Boundary and Conclusion

This paper retains four conditional calculations, none of which closes a physical problem. First, the κ expression is a rearrangement of the Paper I calibration equation. Second, the knot classifications are standard topology, but their physical application and energy ordering have not been derived for CDFT. Third, the displayed ρ_0 is back-solved from an observed mass fit, so it is not a vacuum-density determination. Fourth, the stationary condition for the assumed chiral cosine is calculus conditional on an ansatz; the ansatz, phase, and coupling are not independently derived.

The open requirements are an independently specified physical energy functional, a dimensional closure for its parameters, a justified relation to any soliton model, and a prespecified discriminating observation. Until these exist, this paper must not be cited as a derivation of α, lepton masses, vacuum density, trefoil selection, or a universal knot hierarchy.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

Acknowledgments

The author used AI-assisted tools for formatting checks and code-verification support. The scientific claims, interpretations, and final manuscript decisions are the author's responsibility.

The author is indebted to the foundational work of L. Faddeev and A.J. Niemi [2] (knots and particles), T.H.R. Skyrme [6] (topological solitons), A.F. Rañada [4] (knotted field solutions), G.E. Volovik [7] (vacuum as condensate), and D. Rolfsen [5] (torus knot classification) — whose work provides the mathematical and physical foundations for the topological arguments in this paper.

Declarations

Funding

This research received no external institutional funding.

Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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